EXP: 4 USE CASE DIAGRAM

AIM: To design and develop a Use Case Diagram for the proposed system "Precision Disaster Recovery (PDR-Sync)" and represent system interactions using standard UML notation.

THEORY:

A Use Case Diagram is a type of UML (Unified Modeling Language) diagram used to represent the interaction between users (actors) and a system.

It helps in:

- Identifying system functionalities
- Understanding user interactions
- Visualizing system boundaries
- Supporting requirement analysis

Elements of Use Case Diagram:

- Actor: Represents a user or external system interacting with the system.
  Example: Field Responder, Coordinator, Admin
- Use Case: Represents a functionality or service provided by the system.
  Example: Submit Report, Sync Data
- System Boundary: Defines the scope of the system.
- Relationships

  <<include>> → Mandatory functionality

  <<extend>> → Optional functionality

  Association → Interaction between actor and system

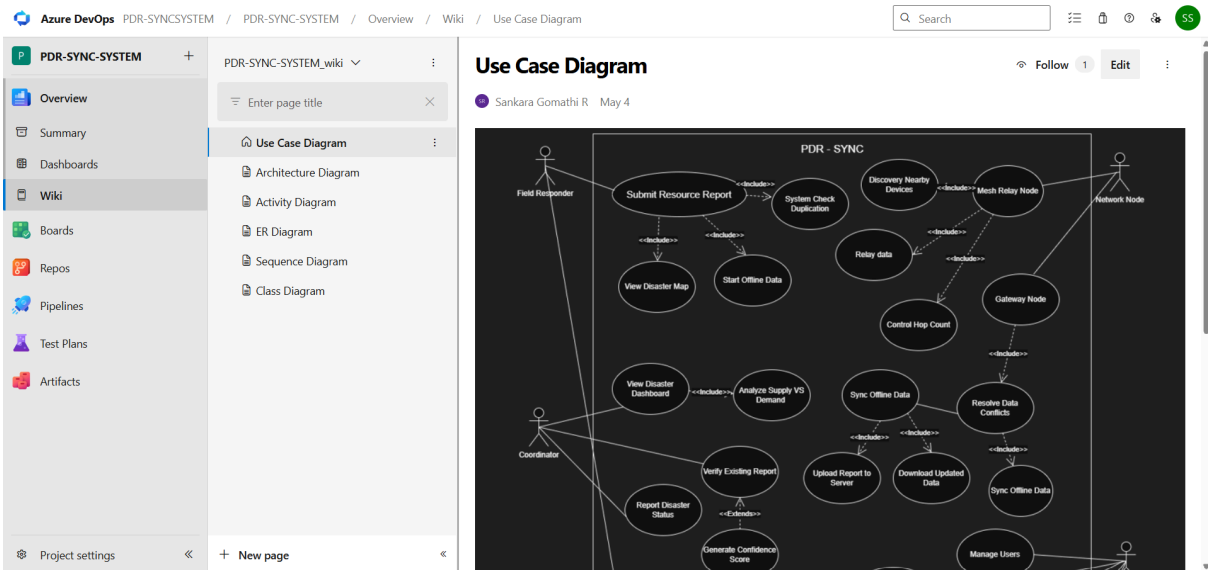
NOTATIONS:

Notation	Description
Stick Figure	Actor
Oval Shape	Use Case

Rectangle	System Boundary
Arrow Line	Association
<<include>>	Mandatory relationship
<<extend>>	Optional relationship

PROCEDURE:

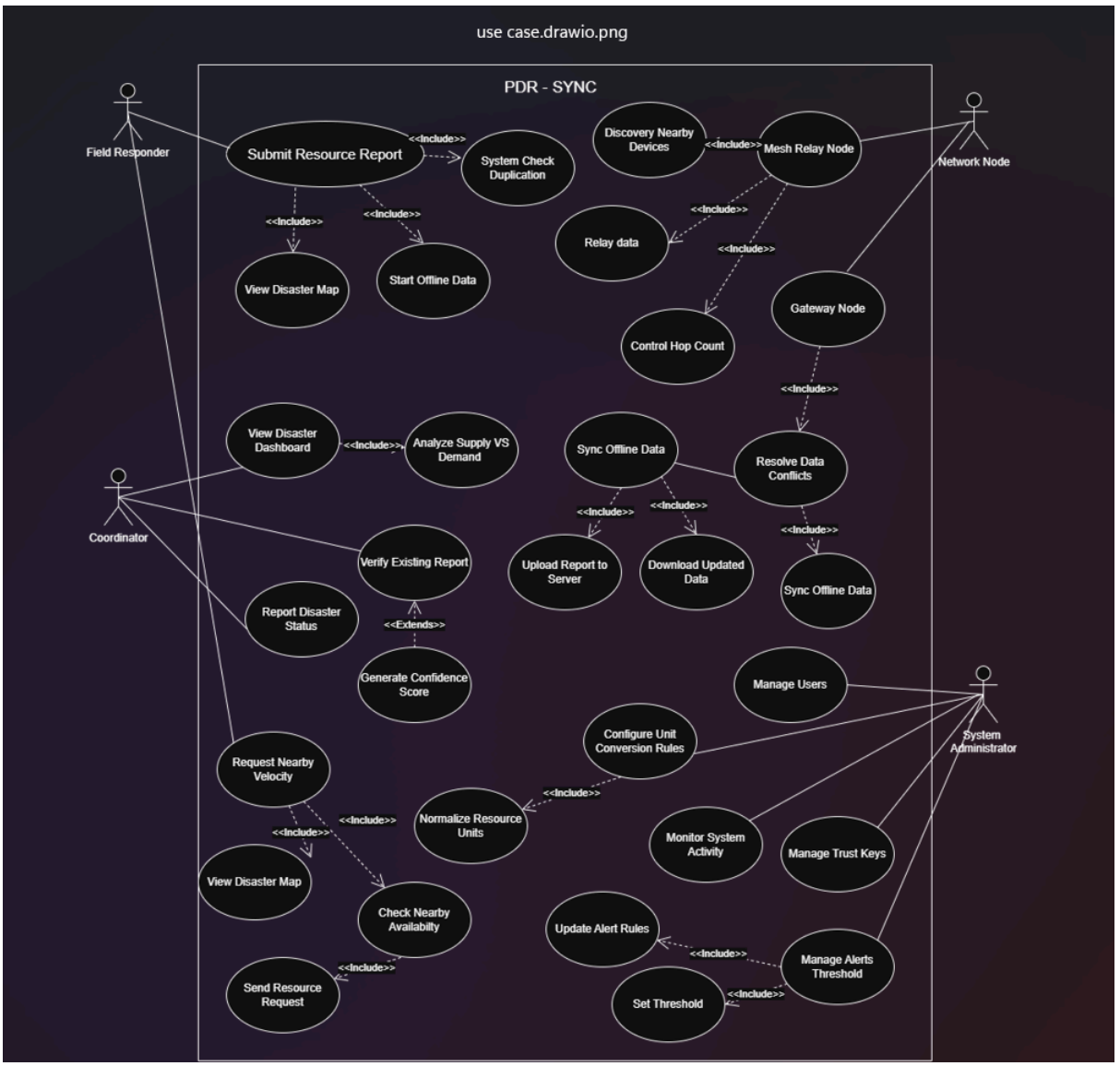
1. Identify all actors interacting with the system
   (Field Responder, Coordinator, Network Node, Administrator)
2. Identify key functionalities (use cases) such as:
   - Resource reporting
   - Disaster status updates
   - Sync operations
   - Data normalization
3. Define relationships between use cases using
   <<include>> and <<extend>>
4. Draw system boundary and place all use cases inside it
5. Connect actors to relevant use cases
6. Validate diagram based on functional requirements



Steps to Create Use Case Diagram in Azure

1. Login to Azure Portal
2. Open Azure DevOps Project
3. Navigate to Boards / Wiki / Repos

4. Use diagram tools such as:
    - draw.io (diagrams.net)
    - Lucidchart
    - or integrate diagram images into Azure Wiki
5. Create a new diagram:
    - Add actors
    - Add use cases
    - Connect relationships
6. Export diagram as image
7. Upload diagram into Azure DevOps project



Use Case Diagram

- Actors: Field Responder, Coordinator, Network Node, Admin

- Core modules: Sync, Mesh Relay, Normalization, Dashboard
- Advanced logic: Conflict Resolution, Confidence Score

Explanation of Diagram

The system consists of multiple actors:

- Field Responder
    - Submits resource reports
    - Views disaster map
    - Works in offline mode
- Coordinator
    - Verifies reports
    - Analyzes supply vs demand
    - Manages disaster status
- Network Node
    - Relays data using mesh networking
    - Supports communication between devices
- System Administrator
    - Manages users
    - Configures system rules
    - Monitors system performance

Key functionalities include:

- Offline data synchronization
- Mesh relay communication
- Conflict resolution
- Data normalization
- Confidence scoring

SCREENSHOT:

RESULT:

Thus, the Use Case Diagram for the system *"Precision Disaster Recovery (PDR-Sync)"* was successfully designed and implemented using UML standards, representing all actors and system functionalities.